

EBU6304 Software Engineering Group Project

2025-26

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First report

1.Requirements

1.1 Fact-finding Techniques

To accurately capture the core needs of target users (applicants and module organizers) for the TA recruitment management system, our group adopted a combination of three fact-finding techniques to ensure comprehensiveness and authenticity of requirements.

1.1.1 Online Searching

We searched for complaints and feedback about the existing educational administration system in online forums and comment sections of related educational videos. Compared with formal interviews or inquiries, people are more relaxed in online environments and more willing to express their true feelings instead of perfunctory responses. For example, some module organizers hope to simplify the approval process for TA-related matters to complete approvals quickly and avoid work delays.

1.1.2 Stakeholder Interviews

Our group members conducted two-way communication with key stakeholders: we sent emails to each course teacher to solicit their opinions on the existing TA management system, and also held exchanges with current and former TAs to discuss the functions, advantages and disadvantages of current TA recruitment methods, areas for improvement, and practical issues in the process. The feedback we collected was the most direct; some teachers had strong reactions, and these are the key issues we need to focus on solving. For example, being able to quickly know who enrolled in his course is an urgent need for our module's module organizer—this is a basic and reasonable requirement. Meanwhile, TAs pointed out that unclear vacancy information and tedious application steps have increased their application burden, which guided us to refine relevant user stories.

1.1.3 Document Analysis

We first analyzed existing materials related to QMUL's TA recruitment process, including historical recruitment notices and manual application forms. This helped us identify pain points in the traditional process, such as repeated input of personal information, unclear vacancy information, and inefficient email-based CV submission. For example, the lack of real-time status tracking for applications led to frequent inquiries from applicants, which laid the foundation for defining user stories like "Application Status Tracking" (TA_08).

1.2 Iteration (Sprint) Plan

Based on the Agile development methodology, we divided the project into 4 sprints, with clear goals, deliverables, and timeframes aligned with the user story priorities and estimation:

Sprint Number	Duration	Core Goals	Corresponding User Stories	Deliverables
Sprint 1	Week 1-2	Complete core functions for all roles (TA/module organizer/Admin) to ensure basic system operation	Must-have stories: TA_01 (Personal Profile Management), TA_02 (Vacancy Browsing & Filtering), TA_04 (Self-Service Password Modification (Logged-in)), TA_05 (Forgot Password - Online Application) MO_01 (Manage TA's Work), MO_02 (View Questionnaire Submissions), MO_03 (User Registration & Role Selection), MO_08 (Self-Service Password Modification (Logged-in)), MO_09 (Forgot Password - Online Application) AD_01 (Monitor Overall TA Workload), AD_02 (Manage System Users), AD_03(Oversee Job Postings), AD_10(Admin Password Reset Processing)	1. Functional prototypes of core modules (TA/module organizer/Admin) 2. User role & vacancy data models design 3. Basic UI framework with role-based access
Sprint 2	Week 3-4	Implement supplementary functions to improve efficiency for all roles	Should-have stories: TA_03 (Integrated Submission), TA_06 (Forgot Password - Direct Admin Contact), TA_08 (Document Preview Before Upload) MO_04 (Simplify Approval Process), MO_05 (Optimize UI Design), MO_06 (Applicant CV Review), MO_07 (Applicant Selection & Approval), MO_10 (Forgot Password	1. Document preview & integrated upload module 2. Streamlined approval workflow (with notifications) 3. Data analysis

			- Direct Admin Contact) AD_04 (View Basic Application Stats), AD_05 (Export Core System Data), AD_11 (Search Users/Jobs), AD_13 (View Individual User Details)	toolkit & optimized UI components
Sprint 3	Week 5-6	Add optional functions to enhance system flexibility & user experience	Could-have stories: TA_07 (Application Deadline Reminder), TA_09 (Save as Draft), TA_10 (Application Status Tracking), TA_11 (Application Withdrawal) AD_06 (View System Key Event Log) , AD_07 (Send Bulk Notifications (Simulated)) , AD_12 (View Job Application Status Summary)	1. UX enhancement suite(reminders/drafts/withdrawal) 2. Security modules (management/logging)
Sprint 4	Week 7-8	Implement advanced optimization and system integration	Want-to-have story: TA_10 (Preference Selection) MO_11 (Support Different Course Types) AD_08 (Configure Recruitment Cycle Dates), AD_09 (Assign Basic User Roles)	1. Advanced modules (preference matching) 2. System integration toolkit (backup/configuration) 3. End-to-end test report & comprehensive docs

1.3 Prioritisation Method

We adopted the **MoSCoW Method** to prioritize user stories, ensuring that critical needs are addressed first while balancing feasibility and user value. The prioritization criteria include: business necessity, user pain point severity, and technical implementation difficulty.

Priority Level	Definition	Distribution of User Stories	Rationale
Must-have	Core functions without which the system cannot operate	TA_01, TA_02, TA_04, TA_05, MO_01, MO_02, MO_03, MO_08,	These functions cover basic user access (registration, password reset), core service scenarios (vacancy browsing, TA management), and are

		MO_09, AD_01, AD_02, AD_03, AD_10	prerequisites for subsequent functions.
Should-have	Important functions that significantly improve user experience but are not mandatory	TA_03, TA_06, TA_08, MO_04, MO_05, MO_06, MO_07, MO_10, AD_04, AD_05, AD_11, AD_13	These functions address key pain points (e.g., integrated CV submission replaces inefficient email transmission) and align with stakeholder feedback, with moderate technical difficulty.
Could-have	Optional Optional functions that enhance flexibility but can be deferred	TA_07, TA_09, TA_10, TA_11, AD_06, AD_07, AD_12	These functions improve user experience (e.g., draft saving) but are not critical to the core process; they can be added after the basic system is stable.
Want-to-have	Nice-to-have functions with low priority and high implementation cost	TA_10, MO_11, AD_08, AD_09	Preference selection requires complex matching logic and is not urgent for initial launch; it can be implemented in the final sprint if resources allow.

1.4 Estimation Methods

To ensure a realistic development schedule and maintain team velocity, we employed a formal **Relative Sizing** approach rather than absolute time estimation. This methodology minimizes the "planning fallacy" by focusing on the complexity, effort, and risks associated with each User Story.

1.4.1 The Fibonacci Scale and Story Points

We adopted the **Fibonacci Sequence (1, 2, 3, 5, 8, 13)** to assign Story Points. The non-linear nature of this scale is crucial: as the complexity of a task increases, the uncertainty grows proportionally.

- **Smaller values (1-5):** Represent well-defined, low-risk tasks.
- **Larger values (8-13):** Signal "Epics" or stories that require decomposition due to high technical uncertainty or extensive data handling.

1.4.2 Triangulation and Reference Stories

To maintain consistency, the team established **Reference Stories** (baselines) to "triangulate" the size of new requirements:

- **Baseline (3 Points):** We defined a standard "CRUD" operation—such as AD_13: View Individual User Details or TA_02: Vacancy Browsing—as a 3-point task. These involve straightforward data queries and basic UI rendering with minimal logic.

- **Comparative Scaling:** Stories with more "moving parts" were scaled upward. For example, TA_10: Application Status Tracking (5 points) was rated higher than the baseline because it requires multi-step status workflow design, real-time notification triggers, and cross-module data synchronization between applicant dashboards and professor review interfaces.

1.4.3 Planning Poker Protocol

To avoid "anchoring bias" (where junior members follow the lead of senior developers), we conducted **Planning Poker** sessions for each story in the backlog:

1. **Independent Assessment:** Each team member privately selected a card representing their estimate.
2. **Revealing & Outlier Discussion:** If estimates differed significantly (e.g., one member voted 3 and another 8), the "outliers" explained their reasoning.

Example Case: For TA_03: Integrated Submission, the initial range was 5 to 13. The developer who voted 13 pointed out the risks of file type validation (.pdf/.docx) and the complexity of linking binary storage to the SQL database. After discussion, the team reached a consensus of 8 points, acknowledging the technical overhead while streamlining the UI requirements.

3. **Consensus:** The final point was only assigned once the whole team agreed on the story's scope and acceptance criteria.

1.4.4 Velocity and Iteration Capacity

By aggregating these points, we calculated the total "Weight" of each Sprint.

- **Sprint 1 (67 points):** Focuses on stable, medium-complexity core features to establish our initial Velocity.
- **Sprint 2 (45 points):** Leveraging the infrastructure built in Sprint 1, we plan to handle higher-complexity tasks.
- **Sprint 3 (34 points):** Focuses on administrative management tools and "Could-have" features to refine the overall user experience and oversight capabilities.
- **Sprint 4 (18 points):** Dedicated to "Hardening," including final bug fixing, security auditing, and documentation to ensure the system meets the "Definition of Done" for final delivery.

This aggregation allows us to monitor if the team is over-committed or under-utilized, ensuring a sustainable pace as mandated by Agile principles.

2.Supporting material

2.1 TA Side

2.1.1 Hands-on Experience

The teaching assistant sent our team members the link for using this system from the TA side, and our members actually tried it out and summarized some areas for optimization. The website is "<https://forms.office.com/e/9B2VdDqmnz>". The interface is as below.



JP/JEI Teaching Assistant (TA) Recruitment - Semester 2 2025-26

Please note: You **do not** have to be a former student from the International School to apply for this work. We welcome applicants from **all** Schools. You **must** be available for the whole of the Semester (the end of February to the end of June).

Please complete this form to apply to be a TA for the Queen Mary University of London modules at BUPT. Please complete this form **only ONCE**. If you make a mistake, you can **edit your form** after submission. You will need to email your CV/Résumé in English, and details can be found at the end of this form. **Please remember to click "Submit" once completed.**

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

* Required

1. Have you been a TA for the Joint Programme(JP)/Joint Education Institute (JEI) in the past? *

☐ Yes

☐ No

You can print a copy of your answer after you submit

Submit

Figure 1: Screenshot of the TA Recruitment Form Used in Hands-on Experience

2.1.2 Interview

We conducted face-to-face interviews with our teaching assistants during tutorial sessions to gather firsthand feedback on the current TA recruitment process. We documented their concerns **in Chinese**, focusing on pain points and optimization suggestions. Key insights include:

- The current process is fragmented: applicants must fill in personal information via a Microsoft Forms questionnaire and separately submit a PDF CV through another channel, leading to redundant work and potential mismatches between application data and resumes.
- There is a lack of transparent progress tracking: after submission, applicants can only wait for email notifications from module organizers, with no way to check the review status in real time, causing uncertainty and anxiety.
- The quota allocation for each course is determined solely by module organizers, with no visible mechanism for applicants to understand the selection criteria or available positions.

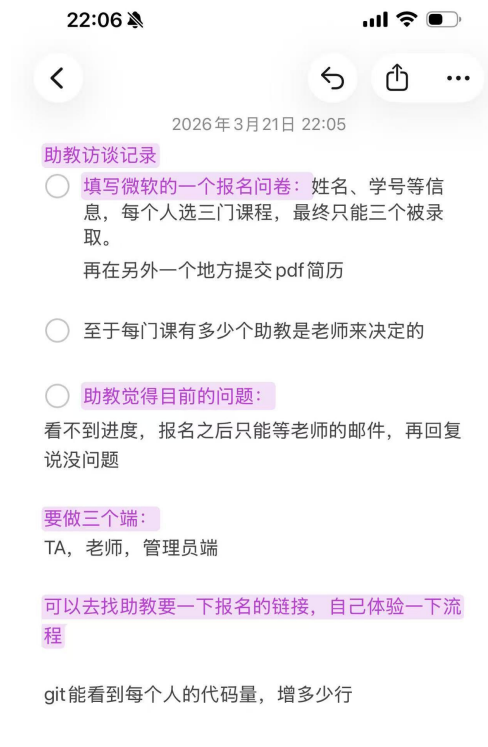


Figure 2: Memo of TA Interview Records (Chinese)

2.2 Module Organizer Side

2.2.1 Email

We conducted email-based communication with our module organiser (module organizer) to collect feedback on the current TA recruitment and management workflow. We sent a formal inquiry email to solicit their pain points and dissatisfaction with the existing process, and received a detailed response outlining key inefficiencies. These insights directly informed the design of our system’s core functions, such as applicant-CV association, status tracking, and batch notification features.

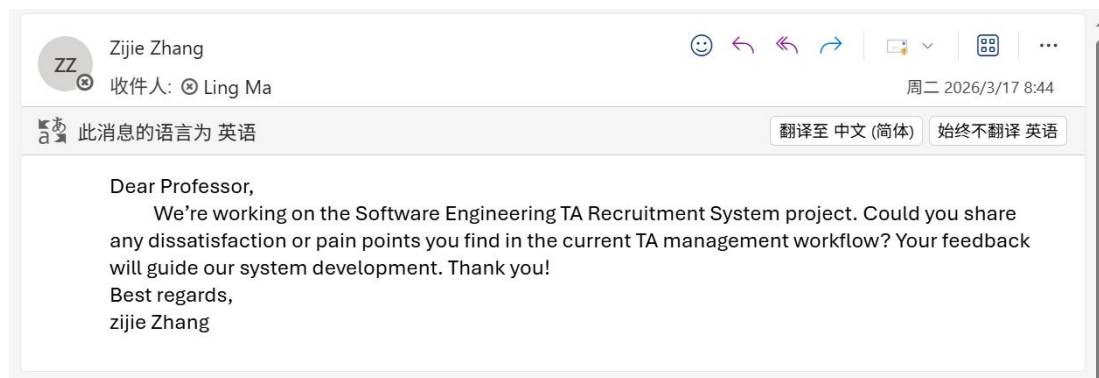


Figure 3: Inquiry Email Sent to module organizer Regarding TA Recruitment Workflow



Figure 4: Module Organizer's Feedback Email on Current TA Management Inefficiencies

2.2.2 Online Search

We have collected some teachers' complaints about this teaching assistant management system from online forums.

Some comments are as follows:

- A. The system is such a headache. Half the time I can't even submit my hours correctly, and the interface is so clunky I have to click through five pages just to do one simple thing. Who designed this?!
- B. Every semester I dread filling out TA info in this system. It's slow, confusing, and never works on the first try. Support never helps either. Just let us use a simple spreadsheet — it would be more reliable!

2.3 Administrator Side

2.3.1 Dashboard Observation

To validate administrator-side requirements, we conducted lightweight observations of 1-2 existing educational recruitment admin dashboards. We analyzed the core functions and workflow design of these platforms, extracting common administrative patterns (e.g., user management, data reporting, and process oversight) to inform our system's admin module design.

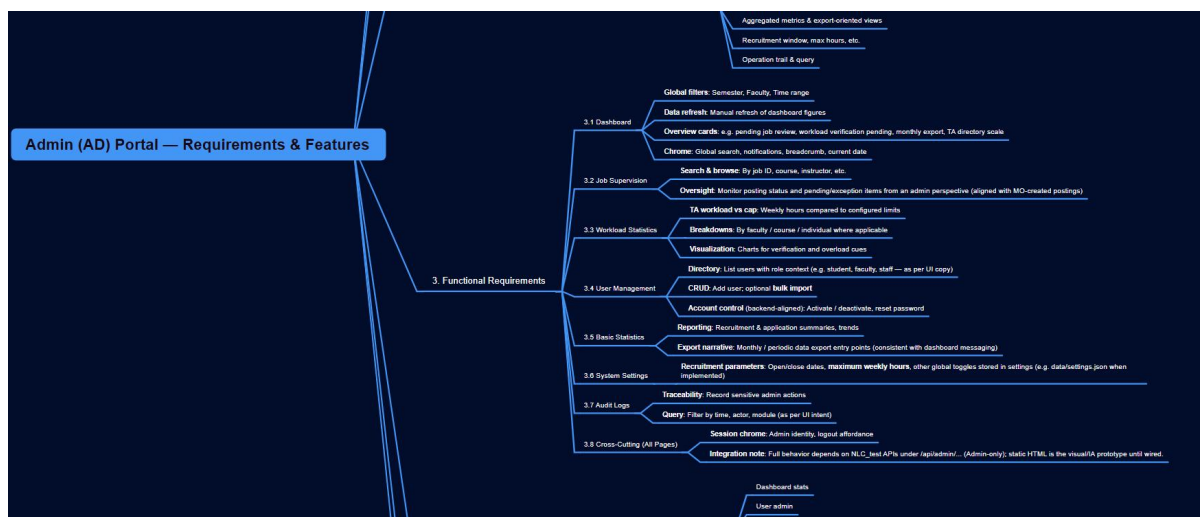


Figure 5: Admin Portal Requirements & Features Mind Map

2.3.2 Questionnaire Survey

We designed a targeted questionnaire via the Questionnaire Star platform and distributed it to teachers and students to collect specific requirements for the administrator role. Key questions included:

1. **What core administrative functions do you believe are essential for a TA recruitment management system? (Select all that apply and briefly explain)**
 - User account & role management
 - Vacancy publishing & quota management
 - Application eligibility review
 - Recruitment data reporting & analysis
 - System notification & reminder sending
 - Data backup & recovery
2. **Which function would most improve the efficiency of the TA recruitment process?**
 - One-click generation of recruitment progress reports
 - Batch notification of application results
 - Automated eligibility screening for applicants
 - Centralized management of applicant information and CVs
 - Role-based access control to protect sensitive data
3. **How important is tracking and visualizing the entire TA recruitment lifecycle for administrative oversight?**
 - Extremely important (critical for process transparency)

- Important (helps identify bottlenecks)
 - Neutral (no significant impact)
 - Slightly unimportant (can be managed manually)
 - Not important at all (unnecessary for our workflow)
4. **What security or data management features do you consider non-negotiable for the administrator module?**
- Password policy enforcement (e.g., complexity, regular reset)
 - Audit logging of key system operations
 - Role-based permission restrictions
 - Regular automated data backup
 - Encryption of sensitive applicant data
5. **Are there any additional administrative pain points or feature requests not covered above? Please describe them in detail.**

The survey focused on identifying essential administrative functions and their expected value, providing quantitative insights for our system design.